

Swarm

Arturo Silvelo

Try New Roads

Swarm

Swarm

Docker Swarm es la **solución nativa de orquestación** de Docker que permite gestionar un cluster de motores Docker como un **sistema virtual único**.

Elementos:

- **Nodo:** máquina que participa en el clúster.
- **Servicio:** Contenedor y réplicas.
- **Stack:** conjunto de servicios desplegados y gestionados juntos.

Características principales

- **Gestión de cluster integrada** - sin software adicional
- **Escalado automático** - ajusta réplicas según demanda
- **Reconciliación del estado deseado** - auto-reparación
- **Service discovery + Balanceo de carga** - DNS y distribución automática
- **Rolling updates** - despliegues sin downtime con rollback
- **Seguro por defecto** - TLS

Crear un Swarm con DinD

Para este curso usaremos **Docker-in-Docker (DinD)** para simular múltiples nodos en una sola máquina.

- Crear el nodo manager

```
docker run -d --privileged --name manager --network host docker:dind
```

- Crear nodos

```
docker run -d --privileged --name worker-1 --hostname worker-1 docker:dind
docker run -d --privileged --name worker-2 --hostname worker-1 docker:dind
```

- Iniciar swarm

```
docker exec manager docker swarm init --advertise-addr <ip|interface>
```

- Añadir nodos

```
docker exec worker-1 docker swarm join <token> <ip>
docker exec worker-2 docker swarm join <token> <ip>
```

- Comprobar swarm

```
$docker exec manager docker node ls
```

ID	HOSTNAME	STATUS	AVAILABILITY	MANAGER STATUS	ENGINE VERSION
jjjwfp1atvi3ho32puxax1pe *	silvelo	Ready	Active	Leader	28.5.0
gi7u04ihrvr95dn6ei114vyh	worker-1	Ready	Active		28.5.0
pjvz11lrjh0pn5re6mhri83df	worker-2	Ready	Active		28.5.0

Gestionar nodos

- Añadir nodos

```
docker run -d --privileged --name worker-3 --hostname worker-3 docker:dind
docker exec worker-3 docker swarm join <token> <ip>
```

- Comprobar swarm

```
$docker exec manager docker node ls
ID                                HOSTNAME    STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
jijwfj1atvi3ho32puxax1pe *      silvelo    Ready    Active           Leader            28.5.0
gi7u04ihrrv95dn6ei1144vyh      worker-1   Ready    Active           28.5.0
pjvz11lrjh0pn5re6mhri83df      worker-2   Ready    Active           28.5.0
7ujsql2taxx4evm045zg7recc      worker-3   Ready    Active           28.5.0
```

- Eliminar nodo

```
docker exec worker-3 docker swarm leave
```

- Comprobar swarm

```
docker exec manager docker node ls
ID                                HOSTNAME    STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
jijwfj1atvi3ho32puxax1pe *      silvelo    Ready    Active           Leader            28.5.0
gi7u04ihrrv95dn6ei1144vyh      worker-1   Ready    Active           28.5.0
pjvz11lrjh0pn5re6mhri83df      worker-2   Ready    Active           28.5.0
7ujsql2taxx4evm045zg7recc      worker-3   Down    Active           28.5.0
```

- Eliminar completamente

```
docker exec manager docker node rm worker-3
```

- Comprobar swarm

```
$docker exec manager docker node ls
ID                                HOSTNAME    STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
jijwfj1atvi3ho32puxax1pe *      silvelo    Ready    Active           Leader            28.5.0
gi7u04ihrrv95dn6ei1144vyh      worker-1   Ready    Active           28.5.0
pjvz11lrjh0pn5re6mhri83df      worker-2   Ready    Active           28.5.0
```

Gestionar Servicios

- Servicio Básico

```
docker exec manager docker service create --name app-base app-base
```

- Servicio con réplicas

```
docker exec manager docker service create --name app-base-replica --replicas 3 app-base
```

- Servicio con réplicas y puerto

```
docker exec manager docker service create --name app-base-port --replicas 2 --publish 8080:80 app-base
```

- Comprobar contenedores

```
docker exec manager docker service ps web
```

